

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
CURRENT AFFAIRS DECODED
SUMMARY NOTES – 26th July - 1st Aug, 2026

Article : A Growth Story That Needs Women at Work

Female Labour Force Participation Rate (LFPR) measures the share of women who are either working or actively seeking/available for work. It includes employed women and unemployed women actively looking for work, but excludes women outside the labour force, such as those engaged only in unpaid domestic work, full-time education, elderly care, childcare, or those not seeking work.

- Formula: Female LFPR = (Female labour force / Total female population) x 100, usually measured for women aged 15 and above.
- Per the **PLFS Annual Report 2025**, female LFPR in usual status for persons aged 15+ stood at 40.0% in 2025, against a male LFPR of 79.1%; rural female LFPR was higher at 45.9%, showing stronger participation in rural than urban areas.
- The **Worker Population Ratio (WPR)** is the ratio of total workers to total population, expressed as a percentage: WPR = (Total workers / Total population) x 100.

The Core Argument

- A new study estimates a 10-percentage-point increase in India's female Work Participation Rate could add nearly two percentage points to GDP growth, making women's employment one of the strongest drivers of economic growth.
- India's female WPR remains below 30%, among the lowest in the world, comparable to Saudi Arabia, Yemen, and West Asia, and it declined steadily between 1983 and 2018.

Why Women's Work Participation Fell

- In the 1980s and early 1990s, female WPR was high due to agricultural participation, but declined as growth and structural transformation shifted the economy away from agriculture from 2004-05 to 2012, with rural women withdrawing due to mechanisation and reduced demand for manual labour.
- Post-2020 increases in women's work participation coincided with a GDP growth slowdown and COVID-19-induced economic distress; women were compelled to intensify unpaid family labour in subsistence agriculture, a "distress-driven feminisation of agriculture," not genuine empowerment.
- India's recent GDP growth has been concentrated in capital-intensive sectors such as finance, IT, and organised manufacturing that don't absorb large numbers of workers, while women's manufacturing employment actually fell between 2013 and 2019.

Article : Renewed War Erases Russian Oil Discounts

Crude oil is a black liquid found in geological formations, a fossil fuel formed from dead organisms buried under intense heat and pressure. Not all crudes are alike; they differ chiefly in weight, sweetness, and TAN count.

- **Weight:** heavy oil evaporates slowly and contains materials used for products like asphalt; light oil produces a greater percentage of gasoline and diesel and requires less processing.
- **Sulphur content:** sweet crude has very low sulphur, well under 1%; sour crude has as much as 1-2% sulphur.

Regional Benchmark Oils

- **WTI (West Texas Intermediate):** oil produced in the United States, typically

lighter and sweet.

- **Brent**: from the Scottish Brent and Ninian systems in the North Sea, typically lighter and sweet.
- **OPEC**: the Organization of Petroleum Exporting Countries, a collective group of crude oils from Saudi Arabia, Iraq, Kuwait, Iran, Libya, Nigeria, the Republic of the Congo, Equatorial Guinea, Gabon, and Venezuela; Venezuelan crude is among the heaviest and most sulphurous in the world.
- **Urals**: a reference oil brand used for pricing Russian export oil, a mix of heavy, sour Urals and Volga region oil with light oil from Western Siberia.

India's Import Dependence

- India imports over 85% of its crude oil requirements, making energy security central to its foreign and economic policy.
- Diversifying import sources across Russia, Gulf states, the US, and Africa is a key strategy to manage price volatility and supply disruption risk.

Article : Centre Sacks 47 NTA Officials, Tightens Law to Tackle Paper Leaks

The **National Testing Agency (NTA)** is an autonomous body under the Ministry of Education, established in 2017 to conduct entrance examinations for higher education institutions, including NEET-UG for medical admissions and various UGC-NET and engineering entrance tests.

Article : Maintenance and Welfare of Parents and Senior Citizens Act, 2007

- "Children" includes son, daughter, grandson, and granddaughter, but not a minor.
- "Maintenance" includes provision for food, clothing, residence, and medical attendance and treatment.
- "Parent" means father or mother, whether biological, adoptive, or step, whether or not they are a senior citizen.
- "Senior citizen" means any Indian citizen who has attained the age of 60 years or above.
- The Act applies when a senior citizen is unable to maintain themselves from their own earnings or property, placing legal responsibility on children and relatives (legal heirs) to provide financial support, food, clothing, residence, and medical care.
- **Maintenance Tribunals** hear petitions from senior citizens; **Appellate Tribunals** hear appeals against tribunal orders.
- The maximum maintenance allowance shall not exceed ₹10,000 per month.

Article : Education Financing and the Falling Budget Share

The **Gross Enrolment Ratio (GER)** is the total number of students enrolled at a specific level of education, regardless of age, divided by the total population of the official age group for that level, multiplied by 100, a standard indicator of access to education, though it doesn't capture quality or learning outcomes.

- The Education Ministry's share of the Union Budget has sharply declined over the past 12 years, from 4.6% in 2013-14 to 2.5% in 2025-26
- Education falls under the **Concurrent List** in the Seventh Schedule, so both the Centre and state governments can legislate and spend on it.
- The **Kothari Commission (1966)** recommended public expenditure on education reach 6% of GDP, a target India has consistently fallen short of; both Union and State budgetary allocations are tracked against this benchmark.

Article : On Guard - The Dengue Vaccine Must Be Made Affordable to Those at Greatest Risk

Dengue fever is a mosquito-borne viral infection transmitted primarily by *Aedes

aegypti* mosquitoes, endemic across India, especially during and after the monsoon; it can range from mild fever to severe dengue haemorrhagic fever or shock syndrome. The same mosquito also transmits chikungunya, yellow fever, and Zika.

- **QDENG** is a recombinant, live-attenuated, tetravalent dengue vaccine designed to protect against all four dengue virus serotypes: DENV-1, DENV-2, DENV-3, and DENV-4.

Article : Zoopharmacognosy - When Animals Medicate Themselves

- A review by G. Allen et al. in *Scientific Reports* (2026) reports orangutans in the Sebangau forest, Central Kalimantan, Indonesia, mixing saliva with the leaves of a plant botanists call dracaena, applying the mixture to their body to purify the air and relieve muscle pain and joint discomfort while carrying their babies.
- Chimpanzees in Nigeria are known to swallow leaves of *Desmodium gangeticum* ("shalaparni" in India) to rid themselves of skin parasites; ancient Ayurveda practitioners such as Sushruta also used this plant for asthma, haemorrhoids, and typhoid.
- Elephants chew oregano leaves for relief from phlegm; monkeys, sheep, dogs, and cats self-medicate by eating grass to remove parasites such as worms and amoeba.
- Animals also consume clay or soil particles (**geophagy**) to alleviate pain; birds use herbs to protect their nests from invaders.

Article : Sarnath Inscribed as India's 45th UNESCO World Heritage Site

Note: Your draft's breakdown table (Cultural 37 + Natural 7 + Mixed 1 = 45) already reflects the post-Sarnath count correctly, but its header said "(44)" and an earlier line separately said "45 sites." The correct, consistent figure throughout is 45 total: 37 cultural, 7 natural, and 1 mixed.

Sarnath's Significance

Located near Varanasi, Uttar Pradesh, **Sarnath** is where Gautama Buddha delivered his first sermon, the Dharmachakra Pravartana, after attaining enlightenment, making it one of Buddhism's four most sacred sites, alongside Lumbini, Bodhi Gaya, and Kushinagar.

- Houses the Ashokan Pillar with the Lion Capital, India's national emblem, and the Dhamek Stupa, among other ancient Buddhist monuments.

UNESCO World Heritage List

A list of sites of outstanding universal cultural or natural significance, maintained under the 1972 World Heritage Convention, to which India is a signatory. Inscription requires meeting at least one of ten selection criteria related to cultural or natural heritage value, along with authenticity, integrity, and an adequate protection and management system.

- With Sarnath's inclusion, India now has 45 UNESCO World Heritage Sites: 37 cultural, 7 natural, and 1 mixed.
- Other Buddhist heritage inscriptions include the Mahabodhi Temple Complex at Bodhi Gaya and the Buddhist Monuments at Sanchi.

Ashoka and the Spread of Buddhism

Emperor Ashoka, in the 3rd century BCE, played a central role in patronising and spreading Buddhism, erecting pillars and stupas at key sites including Sarnath after his conversion following the Kalinga war.

Article : India's Rice Fields Rank Among the World's Top Methane Sources

Rice in India

Rice is mainly a Kharif crop (sown June-July, harvested Nov-Dec), but is also grown as a Rabi and Zaid crop in Southern and Eastern India. It requires temperatures of 25-35°C,

high humidity, and annual rainfall above 100 cm, growing best in deep, clayey, loamy soils with high water-holding capacity.

- India produced a record 150.18 MT of rice in 2024-25, over 28% of global output, on the largest cropped area in the country, about 51-52 million hectares.
- **West Bengal** is the leading producer (about 15%), followed by Uttar Pradesh (~12%) and Punjab (~10%); Punjab has the highest yield due to full irrigation and mechanisation, followed by Tamil Nadu and Telangana.
- India is the world's largest rice exporter, accounting for nearly 40% of global trade; Basmati goes largely to West Asia, non-Basmati to Africa and Southeast Asia.

The Methane Problem

- Greenhouse gas emissions from paddy fields have doubled since the 1960s, per a new study in **Nature Food**; in 2011-2020, paddy fields worldwide produced around 1.1 billion tonnes of CO₂-equivalent emissions annually, up 90.4% from 1961-1980 levels.
- High methane levels result from flooded fields, continuous irrigation, and heavy nitrogen fertiliser use; paddy fields in eastern India were among the global methane hotspots. About 74% of the last decade's nitrous oxide emissions originated from agricultural practices, specifically nitrogen fertilisers and animal manure.
- India's area under paddy cultivation has increased by around 25% since the 1970s, currently covering 55% of India's rainfed harvested area.
- **Why flooded fields produce methane:** continuous flooding creates low-oxygen, anaerobic conditions in paddy soil, ideal for methane-producing microorganisms that decompose organic matter. Methane is a potent greenhouse gas with a global warming potential many times that of CO₂ over 20 years, though with a shorter atmospheric lifetime; agriculture, especially rice cultivation and livestock, is a major anthropogenic source.

System of Rice Intensification (SRI) / Alternate Wetting and Drying

Water-saving irrigation techniques that periodically dry out paddy fields instead of continuous flooding, reducing methane emissions while maintaining or improving yields. SRI was first developed in Madagascar by **Henri de Laulanie**, and is promoted as one of the best practices for sustainable land and ecosystem management (SLEM) under the World Bank-funded SLEM Project.

- One kilogram of rice production typically needs about 3,000-5,000 litres of water, making rice one of India's most water-intensive crops.
- SRI involves cultivating rice with as much organic manure as possible, starting with young seedlings planted singly at wider spacing in a square pattern, intermittent irrigation keeping soil moist but not inundated, and frequent inter-cultivation with a weeder that actively aerates the soil.
- Benefits: cuts water consumption by almost 40%, increases yields by over 30%, can double farmer income by improving input-use efficiency and productivity, is easily adopted by small and marginal farmers (who make up 83% of farmers), improves tolerance to abiotic stress (drought, heat waves) and biotic stress (pests, disease), and can potentially extend to other crops like sugarcane, wheat, and millet.

Article : What Caused the Tunnel Disaster in Sikkim's Namchi?

The **Geological Survey of India (GSI)**, one of the oldest scientific organisations in the country, is set to celebrate its 175th year of geoscientific legacy.

- Set up in 1851 by Sir Thomas Oldham, primarily to find coal deposits for the railways; it has since grown into a repository of geoscience information for various fields.

- Headquartered in Kolkata, with six regional offices at Lucknow, Jaipur, Nagpur, Hyderabad, Shillong, and Kolkata, and a state unit in every state.
- GSI is presently an attached office of the **Ministry of Mines**.

Article : Another Ship With 4 Indians Targeted at Ukraine Port, MEA Issues Advisory

The Black Sea

The **Black Sea**, also called the Euxine Sea, is one of the major water bodies and a famous inland sea of the world, a marginal sea of the Atlantic Ocean located between Eastern Europe and Western Asia.

- Land boundary: bordered by Ukraine to the north and northwest, Russia and Georgia to the east, Türkiye to the south, and Bulgaria and Romania to the west.

Colour Revolutions

Peaceful, mass protest uprisings in post-Soviet states, characterised by symbolic colours, aiming to replace pro-Moscow regimes with democratic governments; though initially successful in creating pro-Western governments, they often resulted in instability, corruption, and disillusionment.

- **Georgia's Rose Revolution (2003), Ukraine's Orange Revolution (2004), and Kyrgyzstan's Tulip Revolution (2005).**

Euromaidan and the Crimean Crisis

Euromaidan was a wave of demonstrations, civil unrest, and revolution in Ukraine, beginning on the night of 21 November 2013 with public protests demanding closer European integration.

- In February and March 2014, Russia invaded and then annexed Ukraine's Crimean Peninsula, in the relative power vacuum immediately following the Revolution of Dignity, marking the beginning of the Russo-Ukrainian war.
- The **Minsk Agreements** refer to the 2014 and 2015 peace accords aimed at ending the eastern Ukraine conflict, negotiated by Ukraine, Russia, the OSCE (Organization for Security and Cooperation in Europe), and separatist representatives of Donetsk and Luhansk.

Article : PRAXIS - Grazing Planet Rings

PRAXIS (Planetary Rings Autonomous Exploration with In-situ Sampling) is a mission concept NASA has selected for its Innovative Advanced Concepts (NIAC) 2026 Phase I funding.

- The mission will consist of a bio-inspired robotic explorer driven by an AI model, capable of directly sampling particles from planetary ring systems, something no mission has ever attempted before.
- Its selection was guided by open questions flagged as important by NASA's Planetary Science Decadal Survey, including how planetary rings form, how they evolve, and what ring particles are made of.

Article : AI's Next Test - Reaching India's Informal Women Workers

India's AI Governance Guidelines (Key Sutras of AI Governance)

- **Trust is the foundation:** the foundational principle guiding all AI development and deployment in India, built on trust that individuals will use these tools responsibly.
- **People first:** AI systems must strengthen human agency and reflect societal values, with humans retaining meaningful control wherever possible, supported by effective human oversight.
- **Innovation over restraint:** all else equal, responsible innovation should be prioritised over cautionary restraint.

- **Fairness and equity:** AI systems should be designed and evaluated to avoid bias or discrimination, particularly against marginalised communities.
- **Accountability:** AI developers and deployers should remain visible and accountable so India's AI ecosystem progresses on trust.
- **Understandable by design:** though AI systems are probabilistic, they must have clear explanations and disclosures helping users and regulators understand how the system works.
- **Safety, resilience, and sustainability:** AI systems should be designed with safeguards minimising risks of harm, and should be robust and resilient.

AI Prosperity for All

- India has built substantial AI infrastructure in a short period: the IndiaAI Mission, the BHASHINI multilingual platform, the AI Governance Guidelines, and the India AI Impact Summit 2026. **BHASHINI** is India's AI-led language translation platform under Digital India.

Article : How a Political Term Can Majorly Fix Urban India's Water Woes

Non-Revenue Water (NRW) is the volume of water produced but not billed to customers, lost through leaks, theft, or metering inaccuracies, a key inefficiency metric for water utilities globally.

Where Water Comes From

- Indian cities have depended increasingly on distant rivers and groundwater, raising energy bills, fuelling political disputes over allocation, and increasing vulnerability to drought and contamination.
- Within a single term, cities can reduce this dependence by treating rainwater and treated wastewater as mainstream sources: stricter enforcement of rainwater harvesting in large buildings, routing treated wastewater to high-demand non-drinking users, and setting realistic targets for non-potable demand from these sources.

How Efficiently Water Is Moved

- Many urban water providers face high non-revenue water, limited wastewater treatment and reuse, and weak revenue collection, key reasons why supply is intermittent, pressure is low, and expansion into poorer neighbourhoods is delayed.
- Achievable within a single term: clarifying responsibility through a clear mandate and coordination with the State government; measuring and managing losses through district metered areas and sustained leak reduction; and modernising customer-facing services through universal metering, better billing, and accessible grievance redress.

Who Receives What, and Who Pays

- Current approaches to coverage, delivery, and affordability haven't closed the equity gap; poorer households often pay more per litre, through time, health, and informal payments, than those receiving piped water at low flat rates.
- Cities can rethink service benchmarks to guarantee a minimum amount of safe water for every resident, and redesign tariffs and subsidies using lifeline blocks and cross-subsidies from higher consumption to protect basic needs while discouraging waste.

Article : India, Bangladesh Must Not Just Renew the Ganga Treaty, They Should Reimagine It

The **Bhagirathi** is one of the two most important headstreams of the Ganga, meeting the Alaknanda at Devprayag to form the Ganga.

- Flowing for nearly 2,525 km, the Ganga passes through Uttarakhand, Uttar

Pradesh, Bihar, Jharkhand, and West Bengal, forming one of the world's most densely populated basins, accounting for 26.4% of India's geographical area and supporting nearly 50% of India's population.

- **Final drainage:** the Bay of Bengal. The river enters the plains at Haridwar, is joined by major tributaries (Ramganga, Yamuna, Gomti, Ghaghara, Gandak, Kosi), meanders widely across fertile alluvial plains due to a low gradient, and finally slows and splits into distributaries forming the world's largest delta.

Left-Bank Tributaries

- **Ramganga:** originates on the southern slopes of Dudhatoli Hill in Chamoli district, Uttarakhand, flows through the Dun valley of Corbett National Park, and meets the Ganga near Kannauj.
- **Gomti:** extends 900 km through Uttar Pradesh and meets the Ganga at Ghazipur.
- **Ghaghara** (also Karnali or Kauriala): a trans-boundary perennial river originating from the Tibetan plateau near Lake Mansarovar, a major left-bank tributary joining the Ganga at Chhapra, Bihar. The **Sarayu** is a left-bank tributary of the Ghaghara that passes through Ayodhya.
- **Gandak:** formed by the union of the Kali and Trisuli rivers, which rise in the Great Himalayan Range in Nepal; it enters the Ganga opposite Patna at a place called Sonapur, after a winding course of 765 km.
- **Kosi:** its unstable nature is attributed to the heavy silt it carries during the monsoon, earning it the name "The Sorrow of Bihar."

Right-Bank Tributaries

- **Yamuna:** the largest and most important tributary, uniting with the Ganga near Triveni Sangam, Allahabad.
- **Son River:** rises in the Amarkantak Plateau, near the origin of the Narmada, and joins the Ganga in Bihar.
- **Damodar:** rises in the hills of the Chotanagpur plateau, flows through a rift valley, and joins the Hugli River 48 km below Kolkata.

The Ganga Water Sharing Treaty, 1996

An agreement between India and Bangladesh on sharing Ganga waters at Farakka during the dry season, signed on 12 December 1996, designed to remain in force for 30 years, meaning it is set to expire in December 2026.

- It manages sharing of Ganges waters at the **Farakka Barrage**, built in 1975 across the Ganges at Farakka, West Bengal, which diverts water to India's Hooghly River to maintain Calcutta Port's navigability.
- Water sharing is by 10-day periods during the lean season (1 January to 31 May) each year.

Article : To Fix Unemployment, Fix the Economy First

Employment elasticity of growth measures how much employment growth accompanies a given rate of GDP growth. Low employment elasticity means economic growth is not translating into proportionate job creation, a persistent concern in India since the 1991 liberalisation reforms.

Article : Urban Indian Private Schools at Risk of Proximity to Toxic Lead Sites

In high-income countries like the US and UK, poorer localities are more exposed to pollution than wealthier ones; that pattern seems reversed in low- and middle-income countries like India, where lead exposure is impactful enough within a 5 km radius that everyone nearby is equally exposed.

Health Impact

- Lead exists naturally deep in the earth, but no amount of exposure is safe for humans; once absorbed it can stay silently stuck to the bones, mimicking

calcium.

- When mixed with blood, it can cause severe kidney and heart problems and spike blood pressure, leading to sudden heart attacks. In children, lead is a powerful neurotoxin that degrades myelin and hampers brain-cell connections, linked to shortened attention spans and a permanent drop in IQ.

The Paradox of Development and Recycling

- Lead-acid batteries, used in most petrol and diesel motor engines, solar cells, wind turbines, and back-up power systems, contribute 86% of global lead consumption; they are popular for being cheap, rechargeable, and easy to recycle, melting at 327°C, achievable with a backyard blowtorch.
- Developed countries recycle lead under strict, high-cost guidelines; in low- and middle-income countries, weak enforcement and cheap labour push much recycling into backyards. In India, despite over 350 well-equipped recycling units, some 90% of the waste ends up in the informal network, where profits are higher.

About Lead

- A toxic, naturally occurring heavy metal, soft, malleable, with a bluish-white lustre; no safe level of exposure has been identified. The WHO identifies lead as one of 10 chemicals of major public health concern (lead poisoning is also called Plumbism or Saturism).
- Lead-based paint remains a major global source of exposure; the WHO and UNEP's Global Alliance to Eliminate Lead Paint urges countries to implement legal restrictions.

Sources of Exposure

- **Occupational:** battery manufacturing and recycling, mining and smelting, glass and ceramic industries, automobile repair, painting and printing, pottery and plumbing, soldering, and lead pipe/plastic manufacturing.
- **Non-occupational (everyday):** traditional medicines, contaminated cosmetics and spices, household paints and storage batteries, vehicular emissions, industrial effluents contaminating soil, water, and dust, food grown in contaminated areas, food cooked or stored in lead-coated utensils, painted toys, and retained bullet fragments.

Article : Co-Existing With Tigers Outside Reserves

The tiger is the largest cat species in the world, both an **umbrella species** and, when chosen to represent a cause, a **flagship species**.

- **Keystone species:** one whose addition to or loss from an ecosystem causes major changes in at least one other species.
- Countries with the most tigers: India, followed by Russia, Indonesia, Nepal, and Thailand.
- Threats: habitat loss, climate change, human-wildlife conflict, inbreeding, disease, hunting, grazing, mining, infrastructure development, and tiger farms.

Key Tiger Conservation Bodies

- **Project Tiger:** a Centrally Sponsored Scheme of the MoEFCC, launched in 1973 for in-situ conservation of wild tigers; Tiger Reserves are recognised under the Wildlife (Protection) Act, 1972, with the **NTCA** holding overarching supervisory authority under the same Act.
- **Global Tiger Forum (GTF):** the only inter-governmental body working exclusively for wild tiger conservation, a forum of 13 tiger range countries.
- **TX2 / GTF goal:** double wild tiger numbers by 2022; India achieved this in 2018, four years ahead of the deadline.
- **CA|TS (Conservation Assured | Tiger Standards):** a WWF initiative offering criteria for tiger sites to check if their management will lead to successful

conservation.

- **Global Tiger Initiative (GTI):** launched in 2008 as a global alliance of governments, civil society, and conservation and scientific communities to save wild tigers from extinction.

The Success and the New Frontier

- Since Project Tiger's 1973 launch, the protected-area network has expanded from nine reserves covering 18,278 sq km to 58 reserves spanning 84,488 sq km, nearly 2.6% of India's landmass.
- India's tiger population rose from 1,411 in 2006 to 3,682 in 2022, home to over 70% of the world's wild tigers; this success has created a new conservation frontier, the growing share of tigers living outside protected areas.
- Per the NTCA, 35-40% of India's tiger population now lives outside designated Tiger Reserves, prompting the MoEFCC to launch the **Tigers Outside Tiger Reserves (TOTR)** initiative.

Why Tigers Are Dispersing

- As territorial carnivores requiring extensive home ranges, tigers disperse naturally when core habitats reach saturation, moving into human-dominated fringes interwoven with villages, farms, and roads, making interactions with people inevitable.

The TOTR Initiative

Aims to establish a scientifically grounded, landscape-based framework for conserving tigers outside protected areas while safeguarding human lives and livelihoods, shifting from reactive crisis management to proactive, science-driven governance.

- **Pillar 1, Conflict reduction:** reducing human-tiger conflict incidents in forest divisions where tigers range outside formally notified reserves.
- **Pillar 2, Community coexistence:** building mechanisms for local communities to coexist with tigers in shared, non-reserve landscapes, rather than treating their presence purely as an enforcement problem.
- **Coverage:** spans 40 forest divisions across 9 states, reflecting the geographic spread of India's tiger population beyond reserve boundaries.

Tiger Reserves of Kerala and the South

- **Periyar Tiger Reserve:** in the Cardamom Hills of the Western Ghats, one of India's oldest Project Tiger reserves, known for Periyar Lake, formed after the Mullaperiyar Dam was built.
- **Parambikulam Tiger Reserve:** on the Kerala-Tamil Nadu border, forming a continuous forest with Anamalai Tiger Reserve, making it one of southern India's most important elephant and tiger corridors; home to the Kanimara Teak, one of Asia's oldest and largest living teak trees.
- **Anamalai Tiger Reserve** (formerly Indira Gandhi Wildlife Sanctuary): in the Anamalai Hills, sharing its western boundary with Parambikulam; includes the tourist destination Topslip and is among the best habitats for the Lion-tailed Macaque and Nilgiri Tahr in the Western Ghats.
- **Kalakad-Mundanthurai Tiger Reserve (KMTR):** the first Tiger Reserve established in Tamil Nadu, lying within the Agasthyamalai Biosphere Reserve, one of India's richest centres of endemism.
- **Mudumalai Tiger Reserve:** in Tamil Nadu's northwestern corner, part of the Nilgiri Biosphere Reserve, joining seamlessly with Bandipur, Nagarhole, and Wayanad Wildlife Sanctuary to form one of India's largest contiguous tiger landscapes; also noted for the Moyar Valley and elephant corridors.
- **Sathyamangalam Tiger Reserve:** forms the ecological bridge between the Western and Eastern Ghats, connecting Mudumalai and Bandipur in the west with BRT and Cauvery forests in the east; once known as the hideout of sandalwood

smuggler Veerappan.

- **Srivilliputhur-Megamalai Tiger Reserve:** India's 52nd Tiger Reserve, combining the Srivilliputhur Grizzled Giant Squirrel Sanctuary and Megamalai Wildlife Sanctuary, protecting the Grizzled Giant Squirrel, Nilgiri Tahr, and Lion-tailed Macaque while strengthening connectivity with the Periyar landscape.

Article : India's Theatre Reform Needs a Readiness Framework

About the Indian Army

- Came into existence in its present form on 26 January 1950, coinciding with the Constitution's commencement and India's transition to a Republic; its origins trace to 1 April 1895, when the British Indian Army was formally established by unifying the Presidency Armies.
- Active strength of approximately 12.37 lakh personnel, the world's largest all-volunteer army and the second-largest standing army after China, with a reserve strength of about 9.6 lakh.
- Army Headquarters: Integrated Headquarters of the Ministry of Defence (Army), New Delhi. Motto: "Seva Paramo Dharma" (Service Before Self).
- Army Day is celebrated on 15 January, marking General (later Field Marshal) K.M. Cariappa taking over as the first Indian Commander-in-Chief of the Indian Army from General Sir Francis Butcher on 15 January 1949.

Rank Equivalence Across Services

Level	Indian Army	Indian Navy	Indian Air Force
1	Lieutenant	Sub Lieutenant	Flying Officer
2	Captain	Lieutenant	Flight Lieutenant
3	Major	Lieutenant Commander	Squadron Leader
4	Lieutenant Colonel	Commander	Wing Commander
5	Colonel	Captain	Group Captain
6	Brigadier	Commodore	Air Commodore
7	Major General	Rear Admiral	Air Vice Marshal
8	Lieutenant General	Vice Admiral	Air Marshal
9	General (COAS)	Admiral (CNS)	Air Chief Marshal (CAS)

Army Command Structure

Command	Headquarters	Primary responsibility
Northern Command	Udhampur, J&K	J&K, Ladakh, Pakistan & China borders
Western Command	Chandimandir, Haryana	Punjab, Himachal Pradesh, parts of J&K
South Western Command	Jaipur, Rajasthan	Rajasthan & Gujarat front with Pakistan

Southern Command	Pune, Maharashtra	Peninsular India & maritime support
Eastern Command	Kolkata, West Bengal	Northeast, China, Myanmar & Bangladesh borders
Central Command	Lucknow, Uttar Pradesh	Central India, Uttarakhand & reserve role
Army Training Command (ARTRAC)	Shimla, Himachal Pradesh	Training, doctrine, and modernisation

- Each command has several **corps** (field formations for specific operational areas, commanded by a Lieutenant General): **Strike Corps** (offensive operations), **Holding Corps** (defensive operations), and **Mixed Corps** (both roles).

Unit Hierarchy

- A corps comprises multiple **divisions** (infantry, armoured, mountain, artillery, and specialised regiments), each commanded by a Major General; divisions are subdivided into **brigades** (commanded by a Brigadier, typically 3-4 battalions plus support elements).
- A **battalion**, the Army's basic fighting unit, is commanded by a Lieutenant Colonel and has about 800-1,000 soldiers; battalions divide into **companies** (commanded by a Major or Captain, about 120 soldiers), which divide into **platoons** (led by a Lieutenant, about 30 soldiers) and **sections** (the smallest unit, led by a non-commissioned officer, 10 soldiers).

Indian Air Force Commands

Command	Headquarters
Western Air Command	New Delhi
Eastern Air Command	Shillong
Central Air Command	Prayagraj
Southern Air Command	Thiruvananthapuram
South Western Air Command	Gandhinagar
Training Command	Bangalore
Maintenance Command	Nagpur

Theatreisation: The Transition

- India's theatre-isation is nearing completion; the US, UK, and China have already shifted to integrated theatre command structures, since modern conflict favours integrated over single-service models.
- Transforming a single-service structure that has stood for seven decades is difficult; merging 17 single-service commands into three integrated theatres carries three real risks.

Three Real Risks

- First, the knowledge gap at the top:** a theatre commander will command forces from three services, at least two of which he hasn't spent decades honing; in a fast-paced conflict like Operation Sindoor (2025), that gap becomes critical when domain expertise matters most.
- Second, cultural friction:** each service has unique values and cultures; merging

them is difficult and will show up sharpest around careers and rank, and a joint force worried about career progression can dampen active combat motivation.

- **Third, slower decision-making:** adding a Vice Chief of Defence Staff alongside the Chief of Defence Staff adds a layer between theatre commanders and the Raksha Mantri, and an untested structure cannot suddenly switch on under fire.

Key Background

- An **Integrated Theatre Command (ITC)** is a unified structure where Army, Navy, and Air Force assets operate under a single commander for a specific geographical area, for better coordination, faster decisions, and improved combat effectiveness.
- The **Kargil Review Committee (1999)** recommended creating integrated theatre commands and a Chief of Defence Staff (CDS) to improve joint operations after coordination failures during the Kargil War.
- The **Shekatkar Committee (2016)** proposed three integrated theatre commands (Western, Northern, and Southern) to enhance tri-service synergy and resource efficiency.
- The **Inter-Services Organisation (Command, Control and Discipline) Act, 2023** empowers theatre commanders with disciplinary control over all three services and promotes jointness through cross-postings.
- The Navy's three principal commands: **Western Naval Command** (Mumbai), **Eastern Naval Command** (Visakhapatnam), and **Southern Naval Command** (Kochi).
- The **Andaman and Nicobar Command (ANC)** is India's first and only existing integrated theatre command, headquartered at Port Blair, set up in October 2001.
- The **Strategic Forces Command (SFC)**, sometimes called the Strategic Nuclear Command, is part of India's Nuclear Command Authority (NCA), responsible for managing and administering the country's tactical and strategic nuclear weapons stockpile and operationalising the NCA's orders; it is also a tri-forces command.

Article : Private R&D Spending Crosses Government's in FY24

India's spending on R&D surged to 0.83% of GDP in 2021-22, the first time the country crossed the 0.8% threshold since 2009-10, per newly disclosed government data.

- The figures reveal a structural shift: private industry accounted for 45.5% of national R&D spending in 2021-22, rising to 51.8% in 2023-24, the first time businesses have contributed more to India's research effort than all levels of government combined.
- The figures were disclosed by the Department of Science and Technology in response to a Lok Sabha question, forming part of the latest Research & Development Statistics 2025-26, though the full statistical report has not yet been publicly released.

Article : IRDAI Ushers in Reforms to Revamp the Insurance Sector

The **Insurance Regulatory and Development Authority of India (IRDAI)** was established following the recommendations of the **Malhotra Committee (1994)**, which examined the need for structural reforms in the insurance sector.

- An autonomous authority under the Ministry of Finance, headquartered in **Hyderabad**.
- Regulates life insurance companies, general insurance companies (including standalone health insurers), reinsurance companies, the agency channel, and intermediaries such as corporate agents, brokers, third-party administrators, and surveyors/loss assessors.

Article : ISRO's NavIC System Can No Longer Provide Navigation Data

Independently

The **Indian Regional Navigation Satellite System (IRNSS)**, operationally known as **NavIC** (Navigation with Indian Constellation), is India's indigenous satellite navigation system.

- Designed to provide accurate positioning within India and up to 1,500 km beyond its borders (the Primary Service Area); an Extended Service Area covers the region between the primary area and the rectangle bounded by 30°S to 50°N latitude and 30°E to 130°E longitude.

Why NavIC Was Needed

- Conceived to ensure strategic autonomy in navigation services; during the 1999 Kargil conflict, the US denied India access to GPS data, exposing the risks of relying on foreign systems. India approved the NavIC project in 2006.

Satellite Generations

- First generation (IRNSS series):** launched 2013-2018, including IRNSS-1A through 1G plus the replacement IRNSS-1I; IRNSS-1H (2017), meant to replace IRNSS-1A, failed to reach orbit due to a heat shield separation failure. Three satellites sit in geostationary orbit and four in inclined geosynchronous orbit; the ground network includes a control centre, precise timing facility, range and integrity monitoring stations, and two-way ranging stations.
- Second generation (NVS series):** developed for improved reliability and expanded capability, including NVS-01, NVS-02, and the planned NVS-03, NVS-04, and NVS-05.

Global and Regional Navigation Systems

System	Country/Region	Type
GPS	United States	Global
GLONASS	Russia	Global
Galileo	European Union	Global
BeiDou	China	Global
NavIC (IRNSS)	India	Regional
QZSS	Japan	Regional (augments GPS over Japan)

- GPS, GLONASS, and Galileo operate with over 20 satellites in Medium Earth Orbit (~20,000 km); BeiDou uses over 40 satellites across Medium Earth Orbit and Geosynchronous Orbit (~35,000 km); NavIC and QZSS have fewer satellites, mainly in higher geosynchronous orbits, focused on regional rather than global coverage.
- For civil aviation, ISRO and the Airports Authority of India have implemented **GAGAN** (GPS Aided Geo Augmented Navigation), a satellite-based augmentation system for Indian airspace.

Article : The Index of Industrial Production (IIP) and Its Base Year Revision

The **IIP** is published monthly by the National Statistical Office (NSO) under the Ministry of Statistics and Programme Implementation (MoSPI), serving as a reliable barometer for the overall health of the industrial sector.

- The IIP base year has been officially updated from 2011-12 to 2022-23, making it the latest major macroeconomic indicator to be revised in 2026 alongside GDP.

Eight Core Sectors (Original Weighting)

Core industries serve as the structural backbone for all other industrial activities. The

combined weight of these core industries in the broader Index of Industrial Production (IIP) is now 32.88% (under the new 2022-23 base), updated from the previous 40.27% under the 2011-12 series.

Addition of Iron Ore: The revised ICI has expanded the list of core industries from eight to nine by including Iron Ore, recognizing its extensive use in steel manufacturing and its vital contribution to industrial development.

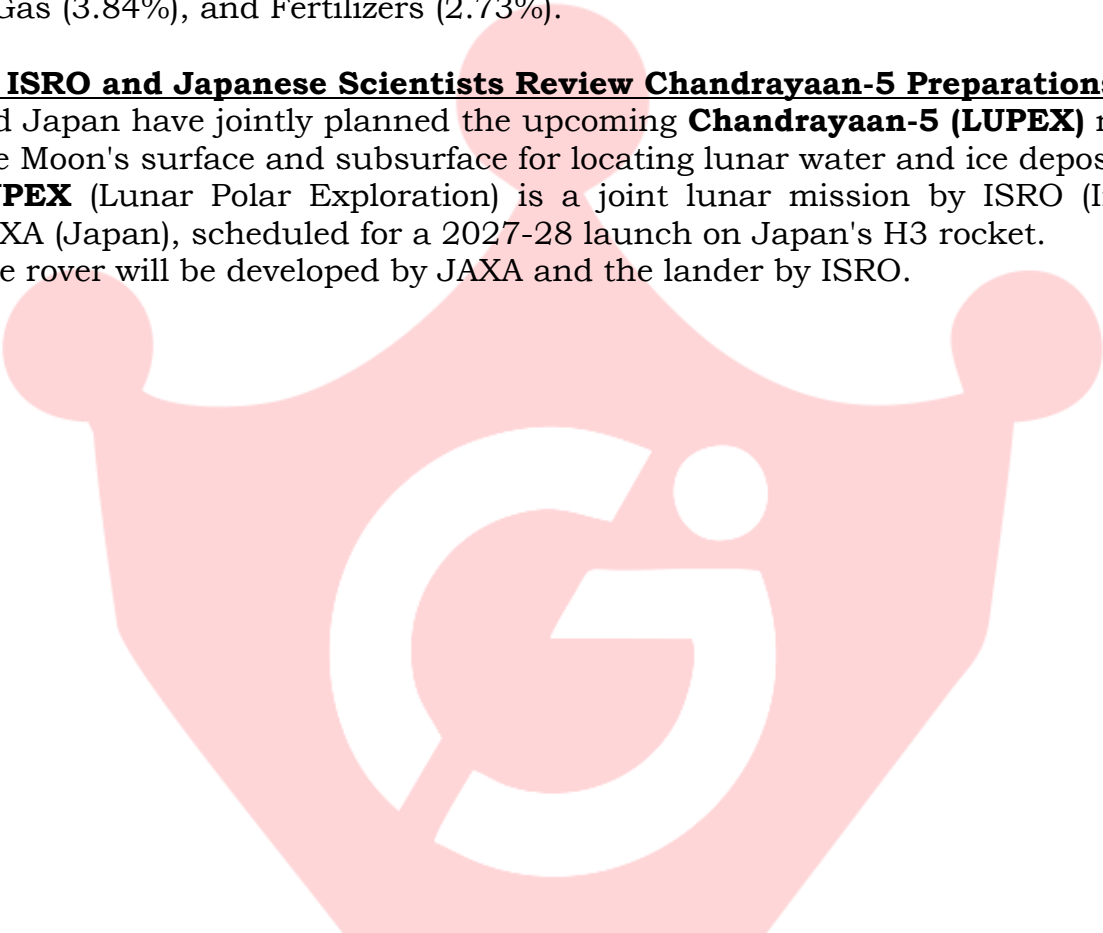
Consequently, the revised ICI now covers Coal, Crude Oil, Natural Gas, Refinery Products, Fertilizers, Steel, Cement, Electricity, and Iron Ore.

Electricity has the highest weight (30.93%), followed by Refinery Products (22.57%), Steel (17.58%), Crude Oil (7.43%), Coal (5.59%), Iron Ore (4.90%), Cement (4.41%), Natural Gas (3.84%), and Fertilizers (2.73%).

Article : ISRO and Japanese Scientists Review Chandrayaan-5 Preparations

India and Japan have jointly planned the upcoming **Chandrayaan-5 (LUPEX)** mission to study the Moon's surface and subsurface for locating lunar water and ice deposits.

- **LUPEX** (Lunar Polar Exploration) is a joint lunar mission by ISRO (India) and JAXA (Japan), scheduled for a 2027-28 launch on Japan's H3 rocket.
- The rover will be developed by JAXA and the lander by ISRO.



GALLANT IAS